

force available, Q starts to slide.

10 B $\Delta U = m\Delta\phi \propto \Delta\phi$

$$\frac{\Delta U_{BC}}{\Delta U_{AB}} = \frac{\Delta\phi_{BC}}{\Delta\phi_{AB}} \Rightarrow \frac{-5.0}{+20} = \frac{\phi_C + 3.0}{-3.0 + 7.0}$$

$$\Rightarrow \phi_C = -4.0 \text{ J kg}^{-1}$$

11 B $\sqrt{2\pi T}$